



APPLICATION FOR MASTER'S-LEVEL AFFILIATED FELLOWSHIPS 2026-2027 COMPETITION

(Application Deadline: 1 December 2025 before 5:00 pm PST)

Applicant Information

Last name of applicant Klukas	First name of applicant Dominic	Initial(s) of middle name(s) H.
UBC student number 64348378	Email klukasdominic@gmail.com	

Degree Information

UBC department / graduate program as of September 2025 (e.g., Chemistry, Asian Studies) Electrical and Computer Engineering	Degree type in which I am registered as of September 2025 (e.g., MSc, MASc, PhD) MASc
Proposed UBC department / graduate program 2026-2027 academic year (e.g., Chemistry, Asian Studies) Electrical and Computer Engineering	Proposed degree type in 2026-2027 academic year (e.g., MSc, MASc, PhD) MASc
Research Subject Area (see Instructions for definitions) Research Subject Area 1, Electrical Engineering	
As of December 31, 2025, I will have completed 3 months of Master's study (total at UBC and other institutions)	
As of December 31, 2025, I will have completed 0 months of doctoral study (total at UBC and other institutions)	

Citizenship

Citizenship Canada
I have applied for 2026-2027 funding from one of the following three federal funding agencies: ☐ CIHR ☒ NSERC ☐ SSHRC ☐ I have not applied for 2026-2027 funding from one of the three federal funding agencies because (provide explanation in the field below):

Signature

I hereby agree that to the terms and conditions outlined in the "Signature" section of the application instructions. Dominic Klukas _____ Applicant's signature Please type your full name above and we will accept that as your signature.	2025-11-27 _____ Date
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Summary of Proposal

Lay Title (99 characters maximum)

Equivariant Reinforcement Learning Policies with Symmetry Breaking

Summary (1,800 characters maximum)

Deep Reinforcement Learning (RL) is an AI algorithm that offers promising solutions to decision and control problems such as robotics tasks. While promising, it typically requires a huge amount of training data and often struggles when the situation changes even a little bit from what it saw during training.

Part of the problem is that standard Deep RL methods don't make use of many of the simple patterns that exist in the real world. For example, many robotics tasks, such as picking up objects and moving them, stay basically the same if the original location of the object is translated or rotated relative to the robot. These patterns are called symmetries, and using them can make learning much faster and more reliable.

When so-called equivariant neural networks are used in Deep RL, the symmetries are built into the algorithm, and automatically respected. However, often symmetries are partially broken in the real world. For example, when picking up an object in one location, we may have to avoid an obstacle not present anywhere else. Because equivariant networks enforce symmetry everywhere, they can't adapt to these exceptions.

In my project I will address this key challenge with three objectives. First, I aim to develop algorithms using mathematical analysis that can detect regions in a state space where symmetries are broken in a robot trajectory dataset. Next, I will design new equivariant neural networks that relax symmetries in specific regions, combining efficient learning when symmetry holds with specialized behavior when needed. Finally, I will classify real robotics tasks and empirically determine when equivariant algorithms, equivariant algorithms with symmetry breaking, and standard Deep RL algorithms perform best.

Sex- and Gender-Based Analysis

Are sex (biological) considerations taken into account in this proposal?

☐ Yes

☒ No

Are gender (socio-cultural) considerations taken into account in this proposal?

☐ Yes

☒ No

If you selected 'yes' to either of the above questions, describe how sex and/or gender considerations will be considered in your program of study. (1,400 characters maximum)

Keywords

List up to 10 keywords that best describe the proposal (40 characters maximum per entry).

1. AI
2. Robotics
3. Reinforcement Learning
4. Geometric Deep Learning
5. Equivariance
6. Symmetries
7.
8.
9.
10.

CRITERIA-BASED AFFILIATED FELLOWSHIPS CHECKLIST

Review the descriptions of the criteria-based awards and indicate those awards for which you are eligible from the list by using the checkboxes.

Note that, in addition to the 'criteria-based' awards noted below, there are numerous 'open/unrestricted' awards that applicants will automatically be considered for.

The award values listed below are subject to change for the 2026/27 academic year.

☐ **#145 - T.K. Lee Scholarship**

A \$1,500 scholarship endowed by T. (Ting) K. Lee is awarded on the recommendation of the Faculty of Graduate and Postdoctoral Studies to a student from the People's Republic of China. Financial circumstances will be considered.

☐ **#146 - Basant Singh and Harnam Kaur Singh Fellowship**

A \$9,400 fellowship is awarded to a full-time student in a Master's or Doctoral program in any discipline, whose work will contribute directly or indirectly to a better understanding and the ultimate advancement of immigrant visible minorities in Canada.

☐ **#334 - Chiu Lin and Grace Ho Memorial Graduate Scholarship**

Scholarships totalling \$27,479 have been made available through an endowment established by an estate gift from Chiu Lin Ho for outstanding international graduate students entering their first year of study in any Faculty. Preference will be given to international students from China, Taiwan or Hong Kong. *Please note: Nominees for this award should be "entering their first year" in September 2025.*

☐ **#350 - Tina and Morris Wagner Foundation Fellowship**

This fund, established through a bequest from Mr. and Mrs. Morris Wagner, provides annual income of \$79,250. The income may be used to provide loans or fellowships for students in the Humanities.

☒ **#484 - Dr. Chi-Kit Wat Scholarship**

Two scholarships of \$3,050 each have been endowed by family and friends in memory of Dr. Chi-Kit Wat. Dr. Wat was associated with the University of British Columbia for many years, both as a student and, later in her career, as a research scientist. One award is made to a visa student from the People's Republic of China who is pursuing post-baccalaureate studies and demonstrates outstanding academic achievement and promise. A second award is made to a Canadian citizen or Permanent Resident of similar academic calibre.

☐ **#486 - C. L. Wang Memorial Scholarship**

A \$1,150 scholarship has been endowed in memory of C.L. Wang by family and friends. C. L. Wang, born Wang Chang-Lin in 1902, was the son of a mandarin official in the last Chinese dynasty before the revolution in 1911. He graduated from Tsing Hua University in Beijing in 1921. With a Tsing Hua scholarship, he studied at the University of Chicago completing his B.Com. in Economics. Following graduation, he worked at the Harris Trust Co. in Chicago, and returned to China in 1926. His career in banking spanned nearly four decades. He was the manager and a Director of the Shanghai Commercial Bank until he retired in 1964. He firmly believed in education and encouraged the young not only in the field of economics and banking to which he devoted his life, but also in engineering and medicine. He died a peaceful death in 1987, at the age of 85. The award is made on the recommendation of the Faculty of Graduate and Postdoctoral Studies to a graduate student from the People's Republic of China, with preference given to a student in Commerce and Business Administration, Engineering or Medicine.

☐ **#527 - Hugo E. Meilicke Memorial Fellowship**



This fund was established by the late Hugo E. Meilicke who gave distinguished service to the community through his business associations and through active participation in organizations such as the Kiwanis Club, the Salvation Army, the Vancouver Foundation, the Crippled Children's Hospital, the Vancouver Art Gallery, and the Vancouver Symphony Society. The annual income of \$44,450 provides awards in various fields such as agriculture, political science, commerce, fine arts, and music.

#1617 - Bank of Montreal Graduate Fellowship

Fellowships totaling \$32,900, endowed by the Bank of Montreal, is available to outstanding graduates in any field but preference is given to graduate students in the field of finance and economics.

☐ **Applying for #1617-finance**

☐ **Applying for #1617-economics**

(students in other fields will be considered automatically)

☐ **#4436 - Werner and Hildegard Hesse Fellowship in Ornithology**

Fellowships totalling \$50,200 have been endowed through a bequest by Werner Hans Hermann Hesse for graduate students engaged in ornithological research projects, with preference given to those studying wild birds. Werner and Hildegard Hesses' passion for wild birds was sparked in night classes on the ecology and conservation of birds at UBC and led them to become leaders in amateur ornithology. The Hesses conducted bird surveys in the Canadian arctic, compiled BC's Christmas Bird Counts for over 20 years, and developed a special concern for the harmful effects of human development on bird habitats and populations. The Hesses delighted in funding ornithological research and, particularly, contributing to the long-term monitoring of population change in seabirds of the Pacific Coast.

☐ **#4811 - Westcoast Energy Inc. Jack Davis Scholarship in Energy Studies**

A \$5,300 scholarship has been endowed by Westcoast Energy Inc. in memory of the Honourable Jack Davis. The award is made to a graduate student in energy studies.

☐ **#4820 - Du Pont Canada Fellowship in Pulp and Paper**

A \$6,300 fellowship has been endowed by Du Pont Canada Inc. and the Province of British Columbia. The award is offered to a graduate student undertaking research related to the pulp and paper industry.

☐ **#4833 - Shaughnessy Hospital Volunteer Society Fellowship in Health Care**

Two \$10,575 fellowships have been endowed by the Shaughnessy Hospital Volunteer Society for graduate students in the field of health care. Preference may be given to students with volunteer experience.

☐ **#4856 - Gertrude Langridge Graduate Scholarship in Humanities**

A scholarship of \$5,600 has been endowed through the bequest of Gertrude Langridge. The award is offered to a graduate student in any field of Humanities.

☐ **#4857 - Gertrude Langridge Graduate Scholarship in Medical Sciences**

A scholarship of \$5,900 has been endowed through the bequest of Gertrude Langridge. The award is offered to a graduate student in any field of Medical Sciences.

☐ **#6326 - C.D. HOWE Graduate Fellowship in Public Policy**

Fellowships totalling \$21,300 fellowship has been endowed by The C.D. Howe Memorial Foundation and The University of British Columbia. The award is offered to a graduate student in public policy.

☐ **#6336 - Anne and John BROWN Fellowship in Diabetes and Obesity Related Research**

Fellowships totalling \$24,000 have been endowed by Ann and John Brown. The awards are offered to graduate students involved in diabetes and obesity related research.

#6350 - Cordula and Gunter Paetzold Fellowship



Fellowships of \$18,000 each have been endowed by Cordula Paetzold and the University of British Columbia for full-time masters or doctoral students, with preference given to graduate students with disabilities, First Nations graduate students, and women students pursuing graduate study in medicine, neuroscience, genetics, health care and epidemiology, pharmaceutical sciences, music, theatre and film, forestry and architecture. If, in any given year, there are not sufficient eligible candidates among the preferred groups, awards may be granted to outstanding graduate students in any field. The fellowships may be renewed for a second year, subject to satisfactory academic standing, attendance and progress.

☐ **#6350 - disability**

☐ **#6350 - First Nations**

☐ **#6350 - woman**

(other eligible students will be considered automatically)

☐ **#6364 - Gerhard Henrik Armauer-Hansen Memorial Scholarship**

Scholarships totalling \$6,450 have been endowed in memory of Dr. Gerhard Henrik Armauer-Hansen, through a bequest by Mary Armauer-Hansen, for graduate students in the Faculty of Medicine or the Department of Microbiology and Immunology who are pursuing research into Hansen's Disease (leprosy) or other bacterial diseases.

☐ **#6391 – Michael W. Stahl Memorial Graduate Scholarship**

A \$1,450 scholarship is offered by Guenter and Diane Stahl in memory of their son, Michael W. Stahl (March 16, 1969 to November 9, 2005), to a graduate student conducting research in the areas of addiction, depression, or mental illness. In spite of some progress from recently initiated treatment for depression, Michael died after many years of struggling with alcohol and drug addiction. The award is made on the recommendation of the Faculty of Graduate and Postdoctoral Studies.

☐ **#6394 - Frieda Granot Graduate Scholarship in Interdisciplinary Research**

A \$400 scholarship has been endowed in honour of Dr. Frieda Granot to recognize over thirty years of service to The University of British Columbia, both as a professor in the Sauder School of Business, starting in 1975, and as Dean of the Faculty of Graduate Studies (1996-2006). The award is made to a Ph.D. student conducting interdisciplinary research.

☐ **#6451 - Pei-Huang Tung and Tan-Wen Tung Graduate Fellowship**

Fellowships totalling \$68,100 have been endowed by Mr. Pei-Huang Tung for international graduate students who are citizens of the People's Republic of China or Taiwan. Fellowships may be renewed for a second year of study at UBC subject to recipients maintaining good academic standing. In adjudicating eligibility, the financial circumstances of eligible candidates may be taken into account.

☐ **#6495 - Andrew Nord Fellowships in Rheumatology**

Fellowships totalling \$73,333 have been endowed through a bequest from the Estate of Andrew Nord. Andrew Nord suffered from the effects of arthritis from the age of 15 and the goal for his bequest is "to prevent the disease from its devastating onslaught, to find better ways to live with the disease, and to find the cause, leading to a cure once and for all." Doctoral students engaged in rheumatology research in any faculty at UBC are eligible to apply for the fellowships.

☐ **#6498 - Elwyn Gregg Memorial Fellowship**

Fellowships totalling \$34,300 have been endowed by a bequest from the Estate of Elwyn Gregg for graduate students in any field of study. Mr. Gregg believed that too often the best graduate talent from UBC may be lost to British Columbia and Canada because of financial attractions and facilities available beyond our borders. It was his hope that this fellowship may be used to attract such abilities to British Columbia. Prominent consideration is to be given to Masters and Ph.D. candidates in Medicine, Languages, Engineering and Forestry, but not necessarily in that order.

☐ **#6503 - Kathleen Simpson Memorial Graduate Scholarship**



A \$2,500 scholarship has been endowed by the Estate of Kathleen Simpson for a graduate student studying social applied science.